Movie Ticket Booking System

Objective:

The objective of this project is to develop a console-based Movie Ticket Booking System using Java. The application allows users to view available movies, select a movie, enter the number of tickets, calculate the total ticket amount, apply discounts, and display booking details. The project demonstrates important Object-Oriented Programming concepts including encapsulation, inheritance, abstraction, polymorphism, interfaces, method overloading, constructors, static members, enums, arrays, exception handling, and control flow statements.

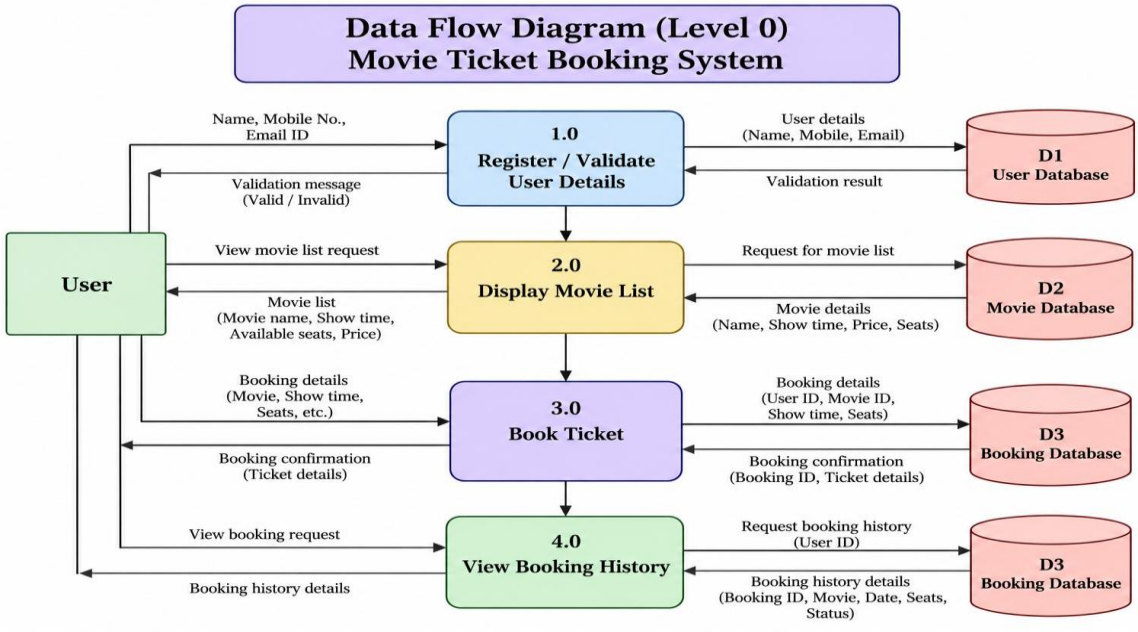
Project Description:

The Movie Ticket Booking System is a console-based Java application designed to simulate a simple real-world movie ticket booking process. The system stores movie information such as movie ID, title, genre, ticket price, and available seats. A customer can select a movie and book tickets. The system calculates the ticket amount, GST, applicable discount, and final amount. The project is organized into separate packages for models, services, and the main application.

Technologies Used:

Programming language used	Java
Application Type	Console-based
Input	Scanner
IDE	Visual Studio Code
Compilation	javac
Execution	java
Packages	model, service, main

Dataflow diagram:



Features :

S.No	Features	Exact file / Class
1	Encapsulation	Movie, Customer, Admin, Booking, Ticket — private fields with getters/setters
2	Data types, constants, scope	Movie.java - int, String, double, static movieCount, final GST_RATE
3	Operators and precedence	Booking.java - ticketAmount + ticketAmount * Movie.GST_RATE - discount
4	Type conversion / casting	MovieTicketApp.java - (int) booking.calculateTotal()
5	Enum	MovieType.java - ACTION, COMEDY, DRAMA, HORROR, SCI_FI
6	Control and jump statements	MovieTicketApp.java, MovieService.java, BookingService.java
7	Array of objects	MovieService.java - Movie[] movies
8	Console I/O / formatted output	MovieTicketApp.java - Scanner; Movie.java / BookingService.java - printf
9	Constructor overloading	Person.java, Customer.java, Admin.java, Movie.java
10	Method overloading	Movie.java - calculatePrice() and calculatePrice(int)
11	Static fields/methods	Movie.java - movieCount/getMovieCount(); Booking.java - bookingCounter
12	this reference	Movie.java, Customer.java, Admin.java, Booking.java
13	String methods	MovieTicketApp.java - trim(), contains(), toLowerCase()
14	Inheritance hierarchy	Person.java → Customer.java and Admin.java
15	super keyword	Customer.java and Admin.java - super() / super(name, phone)
16	Overriding / dynamic binding	Customer.java, Admin.java, MovieTicketApp.java - Person reference to child
17	Abstract class / method	Person.java - abstract class and displayRole()
18	Interface	Payable.java; Ticket.java and Booking.java implement it; BookingService uses interface reference
19	Object method overriding	Movie, Customer, Admin, Ticket, Booking - toString()
20	Final method / class	Ticket.java - final isValidTicketCount() with explanatory comment
21	Custom packages	model, service, main with package declarations and imports

Compilation and Execution:

Compile : javac -d out src/model/*.java src/service/*.java src/main/*.java

Run: java -cp out main.MovieTicketApp

Output:

```
==========================================
      MOVIE TICKET BOOKING SYSTEM
==========================================
Enter your name: Rashmi Hegde
Enter your phone number: 9019116316
Enter your email: hegde853@gmail.com
Welcome, Rashmi Hegde!
Role    : Customer

System Administrator:
Role    : Administrator

============== MAIN MENU ==============
1. View Movies
2. Book Tickets
3. Customer Details
4. Exit
========================================
Enter your choice: 2

================ AVAILABLE MOVIES ================
ID    Movie                    Genre        Price        Seats
--------------------------------------------------
1     Avengers: Endgame        ACTION       Rs.200.00    50
2     Interstellar             SCI_FI       Rs.180.00    40
3     3 Idiots                 COMEDY       Rs.150.00    60
4     The Conjuring            HORROR       Rs.170.00    30
5     Taare Zameen Par         DRAMA        Rs.130.00    45
==================================================

Enter Movie ID: 2
Enter number of tickets: 4
```

```
========== BOOKING DETAILS ==========
Booking ID : 1001
Customer   : Rashmi Hegde
Movie      : Interstellar
Tickets    : 4
Discount   : Rs.0.00
Total      : Rs.849.60
=====================================

Final Amount: Rs.849.60
Rounded amount: Rs.849
```

Conclusion:

The Movie Ticket Booking System successfully demonstrates the application of Java Object-Oriented Programming concepts in a real-world scenario. The project incorporates encapsulation, inheritance, abstraction, polymorphism, interfaces, constructors, method overloading, static members, enums, arrays, control flow, type casting, and package organization. The application was compiled and executed successfully using standard Java commands.